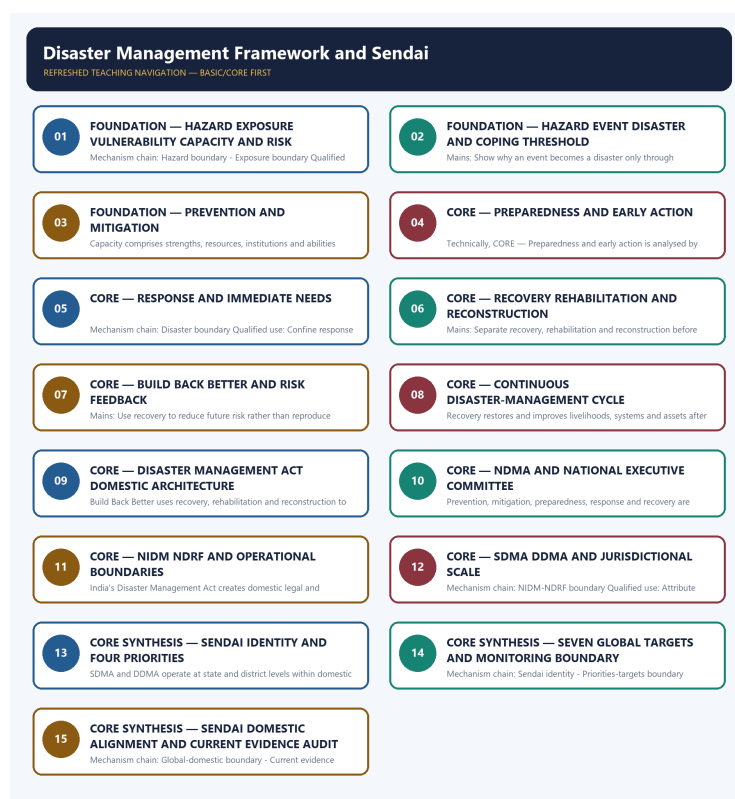


COMPLETE LEARNING SESSION

Disaster Management Framework and Sendai - Learner-v2 Complete Learning Session

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

CONTENTS / SESSION INDEX

Major learning parts and meaningful subtopics are listed in document order. Page numbers are generated from the final PDF layout; PDF bookmarks mirror the same hierarchy.

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Disaster Management Framework and Sendai - Learner-v2 Complete Learning Session

Authoring-only generation: 2026-09-06. No PDF was rendered and no tracker or index was mutated.

SOURCE, PROGRESSION AND CURRENT-LINKAGE AUDIT

- **Generation date:** 2026-09-06.
- **Repository-first evidence:** the Basic owner is taught first and preserved in full; the Advanced owner is retained only in the optional final teaching block.
- **OCR evidence:** Repository Markdown was primary. OCR-searchable local official General Studies papers were used only to confirm printed routed demands. No answer key, marking scheme, unsupported page precision, ecological rate, species count, pollution standard, rule threshold, mission outcome or current status was inferred.
- **Qdrant:** not used; repository Markdown and available OCR context were sufficient.
- **PYQ integrity:** Audited ledgers route the verified 2024 resilience, Sendai-target and urban-flood demands plus related disaster-management questions. No objective key, loss figure, target progress or official model answer is inferred.
- **Live-link boundary:** UNDRR supplied substantive Sendai framework text. Indian official routes did not yield substantive current legal or plan text in this check, so no disaster loss, target progress, indicator, warning coverage, amended power or plan-status claim is asserted.
- **Fact/inference discipline:** no current-affairs item, PYQ wording, figure or quotation is invented.

LIVE OFFICIAL-SOURCE ATTEMPT LOG

The checks below were made on 2026-09-06. Substantive official text is used only for the proposition it supports. Stubs, access failures, unrelated pages and thin landing pages are recorded and are not converted into ecological or current-status claims.

- <https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030> - attempted 2026-09-06; substantive official text confirmed four priorities, seven targets, adoption context and the Build Back Better wording.
- <https://nidm.gov.in/DMAct.asp> - attempted 2026-09-06; the fetched route returned only contact information, so no statutory role, amendment or section claim was imported.
- <https://ndma.gov.in/Governance/DM-Act-2005> - attempted 2026-09-06; the official route returned HTTP 404, so no current Act or institutional wording was imported.
- <https://ndma.gov.in/> - attempted 2026-09-06; no dated plan revision, disaster loss, warning-coverage value or institutional outcome was imported from the general route.

BASIC LEARNING SESSION

DEEP-REVIEW LEARNING CONTRACT

Control	Binding rule for this package
Syllabus boundary	Complete Environment and Ecology Basic/Core is answer-complete before optional Advanced depth.
Ecology boundary	System boundary, scale, trophic level, stock/flow, pool/flux, gross/net, unit and time interval are explicit.
Species boundary	Taxon, range, habitat, population trend, IUCN assessment, Indian legal schedule, CITES/CMS listing and endemism remain distinct.
Law/status boundary	Act, amendment, rule, notification, draft, judgment, policy, target, implementation and observed outcome remain distinct and dated.
Institution boundary	Legal form, parent authority, mandate, jurisdiction, standard, consent, enforcement, science and adjudication are not conflated.
Treaty boundary	Membership, annex/appendix, amendment acceptance, target, COP decision, national instrument and outcome remain distinct.
Climate boundary	Emission flow, concentration stock, cumulative budget, forcing, scenario, baseline, unit, mitigation, adaptation, loss-and-damage, avoidance and removal are exact.
Pollution boundary	Source, emission/load, ambient concentration, exposure, parameter, averaging period, unit, standard and jurisdiction are explicit.
Causal method	Chronology, designation, expenditure, capacity, registration and correlation are not promoted into ecological or policy outcomes without mechanism and evidence.
Practice contract	Every solved item has demand decoding, detailed examiner-grade model, executable timed/compression plan, marks rationale and answer-specific improvement.
Approval	This immutable successor remains approved: false pending explicit approval.

Canonical Basic/Core owner: upsc-ai-kit\knowledge\Environment-and-Ecology\basic\26_Disaster-Management-Framework-and-Sendai.md **Substantive canonical provenance owner:** upsc-ai-kit\knowledge\Environment-and-Ecology\learning-sessions\v2\subject-wide-syllabus\environment-and-ecology-26_Learning-Session.md **Optional Advanced owner:** upsc-ai-kit\knowledge\Environment-and-Ecology\advanced\26_Disaster-Management-Framework-and-Sendai.md **Official syllabus mapping:**

upsc-ai-kit\knowledge\Environment-and-Ecology\OFFICIAL-UPSC-SYLLABUS-MAPPING.md

EVIDENCE, PYQ AND CURRENT-STATUS CONTROL

- Ecological mechanisms retain direction, pool, flux, limiting factor, spatial scale and time scale.
- Current species/news claims retain taxon, source, event/publication date, assessment/listing date and access date.
- IUCN category never substitutes for Wildlife Protection Act schedule, CITES appendix, CMS appendix or endemism.
- Protected-area categories, treaty designations and institution mandates retain exact legal character.
- Acts, amendments, rules, draft instruments, notifications and judgments retain operative status and date.
- Climate figures retain unit, baseline, period, scenario and stock-flow character; global evidence is not silently downscaled to India.
- Mitigation, adaptation and loss-and-damage remain separate; allowance, offset, avoidance, removal, capture and storage remain separate.
- PYQ wording is preserved only where verified; reconstructed or routed demands remain labelled.
- **Current-status note, rechecked 2026-09-06:** volatile targets, standards, schedules, species status, treaty outcomes and programme claims retain source/date/status.

Generation-local live/current sources:

- <https://www.undrr.org/publication/sendai-framework-disaster-risk-reduction-2015-2030> - attempted 2026-09-06; substantive official text confirmed four priorities, seven targets, adoption context and the Build Back Better wording.
- <https://nidm.gov.in/DMAct.asp> - attempted 2026-09-06; the fetched route returned only contact information, so no statutory role, amendment or section claim was imported.
- <https://ndma.gov.in/Governance/DM-Act-2005> - attempted 2026-09-06; the official route returned HTTP 404, so no current Act or institutional wording was imported.
- <https://ndma.gov.in/> - attempted 2026-09-06; no dated plan revision, disaster loss, warning-coverage value or institutional outcome was imported from the general route.

Topic 26 dedicated Disaster Management cross-owners (scope boundary):

- upsc-ai-kit\knowledge\Disaster-Management\basic\01_Concepts-Risk-Resilience-and-Sendai.md
- upsc-ai-kit\knowledge\Disaster-Management\basic\02_Indian-Legal-and-Institutional-Architecture.md
- upsc-ai-kit\knowledge\Disaster-Management\README.md

SESSION 1 - FOUNDATION - Hazard exposure vulnerability capacity and risk

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Hazard exposure vulnerability capacity and risk explains how Hazard boundary and Exposure boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Hazard exposure vulnerability capacity and risk separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Hazard exposure vulnerability capacity and risk must be read through Hazard boundary and Exposure boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Hazard
- exposure
- vulnerability
- capacity
- risk
- boundary

How to use them: Define Hazard, exposure, vulnerability; attach capacity to its source, scale, instrument and status; then qualify the answer with this limit: Do not merge hazard with disaster.

VISUAL FIRST

HAZARD EXPOSURE VULNERABILITY CAPACITY AND RISK

01. Hazard boundary

|
v

02. Exposure boundary

BOUNDARY -> Do not merge hazard with disaster.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

NAMED EVIDENCE AND MECHANISM

- A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
- Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

EXAMINER CAUTION

- Do not merge hazard with disaster.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Define risk through hazard, exposure, vulnerability and capacity.

MINI RECAP

- **Mechanism chain:** Hazard boundary -> Exposure boundary
- **Qualified use:** Define risk through hazard, exposure, vulnerability and capacity.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Hazard exposure vulnerability capacity risk boundary	2. MECHANISM / ARGUMENT connect Hazard boundary and Exposure boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Define risk through hazard, exposure, vulnerability and capacity.	4. UPSC TRAP / ANSWER-USE Do not merge hazard with disaster.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Hazard exposure vulnerability capacity and risk converts a precise environmental distinction into a qualified conclusion			

SESSION 2 - FOUNDATION - Hazard event disaster and coping threshold

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Hazard event disaster and coping threshold explains how Vulnerability boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Hazard event disaster and coping threshold separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Hazard event disaster and coping threshold must be read through Vulnerability boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Hazard
- event
- disaster
- coping
- threshold
- Vulnerability

How to use them: Define Hazard, event, disaster; attach coping to its source, scale, instrument and status; then qualify the answer with this limit: Do not merge exposure with vulnerability.

VISUAL FIRST

HAZARD EVENT DISASTER AND COPING THRESHOLD

01. Vulnerability boundary

BOUNDARY -> Do not merge exposure with vulnerability.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

NAMED EVIDENCE AND MECHANISM

- Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

EXAMINER CAUTION

- Do not merge exposure with vulnerability.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Show why an event becomes a disaster only through disruption and coping limits.

MINI RECAP

- **Mechanism chain:** Vulnerability boundary
- **Qualified use:** Show why an event becomes a disaster only through disruption and coping limits.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Hazard event disaster coping threshold Vulnerability	2. MECHANISM / ARGUMENT connect Vulnerability boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Show why an event becomes a disaster only through disruption and coping limits.	4. UPSC TRAP / ANSWER-USE Do not merge exposure with vulnerability.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Hazard event disaster and coping threshold converts a precise environmental distinction into a qualified conclusion			

SESSION 3 - FOUNDATION - Prevention and mitigation

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Prevention and mitigation explains how Capacity boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Prevention and mitigation separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Prevention and mitigation must be read through Capacity boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Prevention
- mitigation
- Capacity
- boundary
- comprises
- strengths

How to use them: Define Prevention, mitigation, Capacity; attach boundary to its source, scale, instrument and status; then qualify the answer with this limit: Do not omit capacity from a risk analysis.

VISUAL FIRST

PREVENTION AND MITIGATION

01. Capacity boundary

BOUNDARY -> Do not omit capacity from a risk analysis.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

NAMED EVIDENCE AND MECHANISM

- Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

EXAMINER CAUTION

- Do not omit capacity from a risk analysis.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Separate avoiding risk from reducing unavoidable impact.

MINI RECAP

- **Mechanism chain:** Capacity boundary
- **Qualified use:** Separate avoiding risk from reducing unavoidable impact.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS

Prevention | mitigation | Capacity | boundary | comprises | strengths

2. MECHANISM / ARGUMENT

connect Capacity boundary through the source, pathway, instrument and evidence chain

3. CONSEQUENCE / CONTRAST

Separate avoiding risk from reducing unavoidable impact.

4. UPSC TRAP / ANSWER-USE

Do not omit capacity from a risk analysis.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Prevention and mitigation converts a precise environmental distinction into a qualified conclusion

SESSION 4 - CORE - Preparedness and early action

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Preparedness and early action explains how Risk relation fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Preparedness and early action separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Preparedness and early action must be read through Risk relation, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Preparedness
- early
- action
- Risk
- relation
- Disaster

How to use them: Define Preparedness, early, action; attach Risk to its source, scale, instrument and status; then qualify the answer with this limit: Do not infer risk from hazard intensity alone.

VISUAL FIRST

PREPAREDNESS AND EARLY ACTION

01. Risk relation

BOUNDARY -> Do not infer risk from hazard intensity alone.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

NAMED EVIDENCE AND MECHANISM

- Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

EXAMINER CAUTION

- Do not infer risk from hazard intensity alone.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Place warnings, plans, drills and readiness before the event.

MINI RECAP

- **Mechanism chain:** Risk relation
- **Qualified use:** Place warnings, plans, drills and readiness before the event.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Preparedness early action Risk relation Disaster	2. MECHANISM / ARGUMENT connect Risk relation through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Place warnings, plans, drills and readiness before the event.	4. UPSC TRAP / ANSWER-USE Do not infer risk from hazard intensity alone.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Preparedness and early action converts a precise environmental distinction into a qualified conclusion			

SESSION 5 - CORE - Response and immediate needs

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Response and immediate needs explains how Disaster boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Response and immediate needs separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Response and immediate needs must be read through Disaster boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- **Response**
- **immediate**
- **needs**
- **Disaster**
- **boundary**
- **serious**

How to use them: Define Response, immediate, needs; attach Disaster to its source, scale, instrument and status; then qualify the answer with this limit: Do not call every hazard event a disaster.

VISUAL FIRST

RESPONSE AND IMMEDIATE NEEDS

01. Disaster boundary

BOUNDARY -> Do not call every hazard event a disaster.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

NAMED EVIDENCE AND MECHANISM

- A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

EXAMINER CAUTION

- Do not call every hazard event a disaster.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Confine response to immediate life-saving and basic-needs action.

MINI RECAP

- **Mechanism chain:** Disaster boundary
- **Qualified use:** Confine response to immediate life-saving and basic-needs action.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Response immediate needs Disaster boundary serious	2. MECHANISM / ARGUMENT connect Disaster boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Confine response to immediate life-saving and basic-needs action.	4. UPSC TRAP / ANSWER-USE Do not call every hazard event a disaster.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Response and immediate needs converts a precise environmental distinction into a qualified conclusion			

SESSION 6 - CORE - Recovery rehabilitation and reconstruction

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Recovery rehabilitation and reconstruction explains how Prevention-mitigation distinction and Preparedness boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Recovery rehabilitation and reconstruction separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Recovery rehabilitation and reconstruction must be read through Prevention-mitigation distinction and Preparedness boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Recovery
- rehabilitation
- reconstruction
- Prevention-mitigation

- distinction
- Preparedness

How to use them: Define Recovery, rehabilitation, reconstruction; attach Prevention-mitigation to its source, scale, instrument and status; then qualify the answer with this limit: Do not merge prevention with mitigation.

VISUAL FIRST

RECOVERY REHABILITATION AND RECONSTRUCTION

01. Prevention-mitigation distinction

|

v

02. Preparedness boundary

BOUNDARY -> Do not merge prevention with mitigation.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

NAMED EVIDENCE AND MECHANISM

- Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.
- Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

EXAMINER CAUTION

- Do not merge prevention with mitigation.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Separate recovery, rehabilitation and reconstruction before applying Build Back Better.

MINI RECAP

- **Mechanism chain:** Prevention-mitigation distinction -> Preparedness boundary
- **Qualified use:** Separate recovery, rehabilitation and reconstruction before applying Build Back Better.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Recovery rehabilitation reconstruction Prevention-mitigation distinction Preparedness	2. MECHANISM / ARGUMENT connect Prevention-mitigation distinction and Preparedness boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Separate recovery, rehabilitation and reconstruction before applying Build Back Better.	4. UPSC TRAP / ANSWER-USE Do not merge prevention with mitigation.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Recovery rehabilitation and reconstruction converts a precise environmental distinction into a qualified conclusion			

SESSION 7 - CORE - Build Back Better and risk feedback

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Build Back Better and risk feedback explains how Response boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Build Back Better and risk feedback separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Build Back Better and risk feedback must be read through Response boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- **Build**
- **Back**
- **Better**
- **risk**
- **feedback**
- **Response**

How to use them: Define Build, Back, Better; attach risk to its source, scale, instrument and status; then qualify the answer with this limit: Do not call preparedness a response activity.

VISUAL FIRST

BUILD BACK BETTER AND RISK FEEDBACK

01. Response boundary

BOUNDARY -> Do not call preparedness a response activity.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

NAMED EVIDENCE AND MECHANISM

- Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

EXAMINER CAUTION

- Do not call preparedness a response activity.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Use recovery to reduce future risk rather than reproduce vulnerability.

MINI RECAP

- **Mechanism chain:** Response boundary
- **Qualified use:** Use recovery to reduce future risk rather than reproduce vulnerability.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS

Build | Back | Better | risk | feedback | Response

2. MECHANISM / ARGUMENT

connect Response boundary through the source, pathway, instrument and evidence chain

3. CONSEQUENCE / CONTRAST

Use recovery to reduce future risk rather than reproduce vulnerability.

4. UPSC TRAP / ANSWER-USE

Do not call preparedness a response activity.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Build Back Better and risk feedback converts a precise environmental distinction into a qualified conclusion

SESSION 8 - CORE - Continuous disaster-management cycle

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Continuous disaster-management cycle explains how Recovery boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Continuous disaster-management cycle separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Continuous disaster-management cycle must be read through Recovery boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Continuous
- disaster-management
- cycle
- Recovery
- boundary
- restores

How to use them: Define Continuous, disaster-management, cycle; attach Recovery to its source, scale, instrument and status; then qualify the answer with this limit: Do not call emergency response long-term recovery.

VISUAL FIRST

CONTINUOUS DISASTER-MANAGEMENT CYCLE

01. Recovery boundary

BOUNDARY -> Do not call emergency response long-term recovery.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

NAMED EVIDENCE AND MECHANISM

- Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

EXAMINER CAUTION

- Do not call emergency response long-term recovery.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Link every phase through feedback rather than a one-way cycle.

MINI RECAP

- **Mechanism chain:** Recovery boundary
- **Qualified use:** Link every phase through feedback rather than a one-way cycle.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS

Continuous | disaster-management | cycle | Recovery | boundary | restores

2. MECHANISM / ARGUMENT

connect Recovery boundary through the source, pathway, instrument and evidence chain

3. CONSEQUENCE / CONTRAST

Link every phase through feedback rather than a one-way cycle.

4. UPSC TRAP / ANSWER-USE

Do not call emergency response long-term recovery.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Continuous disaster-management cycle converts a precise environmental distinction into a qualified conclusion

SESSION 9 - CORE - Disaster Management Act domestic architecture

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Disaster Management Act domestic architecture explains how Build-Back-Better boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Disaster Management Act domestic architecture separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Disaster Management Act domestic architecture must be read through Build-Back-Better boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Disaster
- Management
- domestic
- architecture
- Build-Back-Better
- boundary

How to use them: Define Disaster, Management, domestic; attach architecture to its source, scale, instrument and status; then qualify the answer with this limit: Do not reduce recovery to rebuilding identical assets.

VISUAL FIRST

DISASTER MANAGEMENT ACT DOMESTIC ARCHITECTURE

01. Build-Back-Better boundary

BOUNDARY -> Do not reduce recovery to rebuilding identical assets.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

NAMED EVIDENCE AND MECHANISM

- Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

EXAMINER CAUTION

- Do not reduce recovery to rebuilding identical assets.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Anchor domestic roles in the Disaster Management Act.

MINI RECAP

- **Mechanism chain:** Build-Back-Better boundary
- **Qualified use:** Anchor domestic roles in the Disaster Management Act.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Disaster Management domestic architecture Build-Back-Better boundary	2. MECHANISM / ARGUMENT connect Build-Back-Better boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Anchor domestic roles in the Disaster Management Act.	4. UPSC TRAP / ANSWER-USE Do not reduce recovery to rebuilding identical assets.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Disaster Management Act domestic architecture converts a precise environmental distinction into a qualified conclusion			

SESSION 10 - CORE - NDMA and National Executive Committee

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: NDMA and National Executive Committee explains how Cycle-continuum boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, NDMA and National Executive Committee separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

NDMA and National Executive Committee must be read through Cycle-continuum boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- **NDMA**
- **National**
- **Executive**
- **Committee**
- **Cycle-continuum**
- **boundary**

How to use them: Define NDMA, National, Executive; attach Committee to its source, scale, instrument and status; then qualify the answer with this limit: Do not make the disaster cycle a one-way sequence.

VISUAL FIRST

NDMA AND NATIONAL EXECUTIVE COMMITTEE

01. Cycle-continuum boundary

BOUNDARY -> Do not make the disaster cycle a one-way sequence.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

NAMED EVIDENCE AND MECHANISM

- Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

EXAMINER CAUTION

- Do not make the disaster cycle a one-way sequence.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Differentiate policy authority from executive coordination.

MINI RECAP

- **Mechanism chain:** Cycle-continuum boundary
- **Qualified use:** Differentiate policy authority from executive coordination.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS NDMA National Executive Committee Cycle-continuum boundary	2. MECHANISM / ARGUMENT connect Cycle-continuum boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Differentiate policy authority from executive coordination.	4. UPSC TRAP / ANSWER-USE Do not make the disaster cycle a one-way sequence.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NDMA and National Executive Committee converts a precise environmental distinction into a qualified conclusion			

SESSION 11 - CORE - NIDM NDRF and operational boundaries

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: NIDM NDRF and operational boundaries explains how Domestic-law boundary and NDMA-NEC boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, NIDM NDRF and operational boundaries separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

NIDM NDRF and operational boundaries must be read through Domestic-law boundary and NDMA-NEC boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- NIDM
- NDRF
- operational
- boundaries

- Domestic-law
- boundary

How to use them: Define NIDM, NDRF, operational; attach boundaries to its source, scale, instrument and status; then qualify the answer with this limit: Do not derive domestic legal powers from Sendai.

VISUAL FIRST

NIDM NDRF AND OPERATIONAL BOUNDARIES

01. Domestic-law boundary

|

v

02. NDMA-NEC boundary

BOUNDARY -> Do not derive domestic legal powers from Sendai.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

NAMED EVIDENCE AND MECHANISM

- India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
- NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

EXAMINER CAUTION

- Do not derive domestic legal powers from Sendai.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Differentiate training and research from specialised response operations.

MINI RECAP

- **Mechanism chain:** Domestic-law boundary -> NDMA-NEC boundary
- **Qualified use:** Differentiate training and research from specialised response operations.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS NIDM NDRF operational boundaries Domestic-law boundary	2. MECHANISM / ARGUMENT connect Domestic-law boundary and NDMA-NEC boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Differentiate training and research from specialised response operations.	4. UPSC TRAP / ANSWER-USE Do not derive domestic legal powers from Sendai.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NIDM NDRF and operational boundaries converts a precise environmental distinction into a qualified conclusion			

SESSION 12 - CORE - SDMA DDMA and jurisdictional scale

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: SDMA DDMA and jurisdictional scale explains how NIDM-NDRF boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, SDMA DDMA and jurisdictional scale separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

SDMA DDMA and jurisdictional scale must be read through NIDM-NDRF boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- SDMA
- DDMA
- jurisdictional
- scale
- NIDM-NDRF
- boundary

How to use them: Define SDMA, DDMA, jurisdictional; attach scale to its source, scale, instrument and status; then qualify the answer with this limit: Do not merge NDMA with NEC.

VISUAL FIRST

SDMA DDMA AND JURISDICTIONAL SCALE
01. NIDM-NDRF boundary
BOUNDARY -> Do not merge NDMA with NEC.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

NAMED EVIDENCE AND MECHANISM

- NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

EXAMINER CAUTION

- Do not merge NDMA with NEC.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Attribute state and district roles only to their owned jurisdiction.

MINI RECAP

- **Mechanism chain:** NIDM-NDRF boundary
- **Qualified use:** Attribute state and district roles only to their owned jurisdiction.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW | SUBTOPIC CLOSURE FLOW

1. KEY TERMS / DEFINITIONS SDMA DDMA jurisdictional scale NIDM-NDRF boundary	2. MECHANISM / ARGUMENT connect NIDM-NDRF boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Attribute state and district roles only to their owned jurisdiction.	4. UPSC TRAP / ANSWER-USE Do not merge NDMA with NEC.
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SDMA DDMA and jurisdictional scale converts a precise environmental distinction into a qualified conclusion

SESSION 13 - CORE SYNTHESIS - Sendai identity and four priorities

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Sendai identity and four priorities explains how State-district boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Sendai identity and four priorities separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Sendai identity and four priorities must be read through State-district boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Sendai
- identity
- four
- priorities
- State-district
- boundary

How to use them: Define Sendai, identity, four; attach priorities to its source, scale, instrument and status; then qualify the answer with this limit: Do not merge NIDM training with NDRF operations.

VISUAL FIRST

SENDAI IDENTITY AND FOUR PRIORITIES

01. State-district boundary

BOUNDARY -> Do not merge NIDM training with NDRF operations.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

NAMED EVIDENCE AND MECHANISM

- SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

EXAMINER CAUTION

- Do not merge NIDM training with NDRF operations.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** State Sendai's global non-binding identity before its priorities.

MINI RECAP

- **Mechanism chain:** State-district boundary
- **Qualified use:** State Sendai's global non-binding identity before its priorities.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Sendai identity four priorities State-district boundary	2. MECHANISM / ARGUMENT connect State-district boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST State Sendai's global non-binding identity before its priorities.	4. UPSC TRAP / ANSWER-USE Do not merge NIDM training with NDRF operations.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Sendai identity and four priorities converts a precise environmental distinction into a qualified conclusion			

SESSION 14 - CORE SYNTHESIS - Seven global targets and monitoring boundary

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Seven global targets and monitoring boundary explains how Sendai identity and Priorities-targets boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Seven global targets and monitoring boundary separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Seven global targets and monitoring boundary must be read through Sendai identity and Priorities-targets boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Seven
- global
- targets
- monitoring
- boundary
- Sendai

How to use them: Define Seven, global, targets; attach monitoring to its source, scale, instrument and status; then qualify the answer with this limit: Do not move national roles automatically to a DDMA.

VISUAL FIRST

SEVEN GLOBAL TARGETS AND MONITORING BOUNDARY

01. Sendai identity

|
v

02. Priorities-targets boundary

BOUNDARY -> Do not move national roles automatically to a DDMA.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

NAMED EVIDENCE AND MECHANISM

- The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.
- Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

EXAMINER CAUTION

- Do not move national roles automatically to a DDMA.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** List global targets separately from action priorities.

MINI RECAP

- **Mechanism chain:** Sendai identity -> Priorities-targets boundary
- **Qualified use:** List global targets separately from action priorities.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Seven global targets monitoring boundary Sendai	2. MECHANISM / ARGUMENT connect Sendai identity and Priorities-targets boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST List global targets separately from action priorities.	4. UPSC TRAP / ANSWER-USE Do not move national roles automatically to a DDMA.
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: Seven global targets and monitoring boundary converts a precise environmental distinction into a qualified conclusion			

SESSION 15 - CORE SYNTHESIS - Sendai domestic alignment and current evidence audit

DEFINITION / WHAT THIS IS CALLED

Plain-language definition: Sendai domestic alignment and current evidence audit explains how Global-domestic boundary and Current evidence boundary fit into one examinable ecological mechanism.

Technical definition: In Environment and Ecology, Sendai domestic alignment and current evidence audit separates the environmental system and receiving medium from the measured parameter, source or precursor, responsible actor, regulatory instrument, spatial scale, temporal stage, rule vintage and legal or scientific status attached to the claim.

ANSWER-GRABBING OPENING - WRITE/ADAPT IN THE EXAM

Sendai domestic alignment and current evidence audit must be read through Global-domestic boundary and Current evidence boundary, with the ecological level, system boundary and measured parameter fixed before the inference.

MUST-WRITE KEYWORDS

- Sendai
- domestic
- alignment
- audit
- Global-domestic
- boundary

How to use them: Define Sendai, domestic, alignment; attach audit to its source, scale, instrument and status; then qualify the answer with this limit: Do not call Sendai a binding treaty.

VISUAL FIRST

SENDAI DOMESTIC ALIGNMENT AND CURRENT EVIDENCE AUDIT

01. Global-domestic boundary

|
v

02. Current evidence boundary

BOUNDARY -> Do not call Sendai a binding treaty.

This topic-specific rail fixes the evidence sequence and its exam boundary before analysis.

CORE EXPLANATION

Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

NAMED EVIDENCE AND MECHANISM

- Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.
- Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

EXAMINER CAUTION

- Do not call Sendai a binding treaty.

EXAM LINK

- **Prelims:** Preserve the exact receiving medium, system boundary, measured parameter, actor, process, spatial and temporal scale, rule vintage and scientific or legal status; never turn a context-dependent pattern into a universal rule.
- **Mains:** Close with dated domestic law, plan, indicator and UNDRR evidence.

MINI RECAP

- **Mechanism chain:** Global-domestic boundary -> Current evidence boundary
- **Qualified use:** Close with dated domestic law, plan, indicator and UNDRR evidence.

CLOSING RECALL FLOW

SUBTOPIC CLOSURE FLOW SUBTOPIC CLOSURE FLOW			
1. KEY TERMS / DEFINITIONS Sendai domestic alignment audit Global-domestic boundary	2. MECHANISM / ARGUMENT connect Global-domestic boundary and Current evidence boundary through the source, pathway, instrument and evidence chain	3. CONSEQUENCE / CONTRAST Close with dated domestic law, plan, indicator and UNDRR evidence.	4. UPSC TRAP / ANSWER-USE Do not call Sendai a binding treaty.

COMPLETE BASIC OWNER EVIDENCE BANK

Subject: Environment and Ecology | **Tier:** Must-Do (foundation) | **GS Paper:** GS-III (Disaster Management) + Prelims. **Core area:** Disaster-risk-reduction governance architecture. **Grounded in:** Disaster Management Act, 2005 (India Code); Sendai Framework for Disaster Risk Reduction 2015-2030 (UNDRR); NDMA guidelines; audited UPSC Environment/Disaster Management PYQs (2024-2026 Prelims and 2024-2025 Mains). **FACT:** = source-grounded | **ANALYSIS:** = analytical linkage | **CURRENT:** = dated current-affairs anchor. **Companion:** advanced/26_Disaster-Management-Framework-and-Sendai.md.

1. Visual foundation

DISASTER MANAGEMENT CYCLE (four linked phases, not a single event response)
MITIGATION -> PREPAREDNESS -> RESPONSE -> RECOVERY -> (feeds back into MITIGATION)

INDIA'S THREE-TIER INSTITUTIONAL STRUCTURE (Disaster Management Act, 2005)
NDMA (National) -> chaired by the Prime Minister
SDMA (State) -> chaired by the Chief Minister
DDMA (District) -> chaired by the District Magistrate/Collector

SENDAI FRAMEWORK (2015-2030) - FOUR PRIORITIES FOR ACTION
1. Understanding disaster risk
2. Strengthening disaster-risk governance to manage risk
3. Investing in disaster-risk reduction for resilience
4. Enhancing disaster preparedness for effective response, and to "Build Back Better" in recovery, rehabilitation and reconstruction

Core proposition: Effective disaster management is a continuous cycle (mitigation, preparedness, response, recovery), not a single reactive event-response - India's Disaster Management Act, 2005 established a three-tier institutional structure (NDMA-SDMA- DDMA) that operationalises this cycle, aligned globally with the Sendai Framework's shift from disaster *management* to disaster *risk reduction* as the guiding philosophy.

2. Essential definitions

Concept	Exam-ready meaning
FACT: Disaster Management Act, 2005	India's principal law establishing the institutional framework (NDMA, SDMA, DDMA) for disaster prevention, mitigation, preparedness, response and recovery.
FACT: National Disaster Management Authority (NDMA)	Apex national body, chaired by the Prime Minister, laying down disaster-management policies and guidelines.
FACT: State Disaster Management Authority (SDMA)	State-level body, chaired by the Chief Minister, responsible for state-level disaster-management planning.
FACT: District Disaster Management Authority (DDMA)	District-level body, chaired by the District Magistrate/Collector, responsible for district-level plan implementation.
FACT: Sendai Framework for Disaster Risk Reduction 2015-2030	A voluntary, non-binding UN framework (successor to the Hyogo Framework for Action), adopted in Sendai, Japan, in 2015, guiding global disaster-risk-reduction efforts.
FACT: "Build Back Better"	A Sendai Framework principle emphasising that post-disaster recovery/reconstruction should reduce future risk and improve resilience, not merely restore pre-disaster conditions.

3. Topic mechanism

- The Disaster Management Act, 2005 (enacted following the 2004 Indian Ocean tsunami's demonstration of gaps in India's disaster-response coordination) created a formal three-tier institutional structure - NDMA at the national level, SDMA at the state level, and DDMA at the district level - each responsible for disaster-management planning and coordination at its respective level.
- The Act mandates a shift from a purely reactive, relief-and-response orientation

(historically dominant in Indian disaster governance) toward a comprehensive approach covering the full cycle: prevention, mitigation, capacity-building, preparedness, response, and rehabilitation/reconstruction.

3. The Sendai Framework (2015-2030), adopted at the UN's Third World Conference on Disaster Risk Reduction in Sendai, Japan, succeeded the earlier Hyogo Framework for Action and articulated four priorities for action: understanding disaster risk; strengthening disaster-risk governance; investing in disaster-risk reduction for resilience; and enhancing preparedness for effective response paired with "Build Back Better" recovery.

4. India, as a Sendai Framework signatory, aligns its National Disaster Management Plan with Sendai's priorities, emphasising risk understanding (hazard mapping, vulnerability assessment), governance strengthening (institutional coordination), and resilience investment (disaster-resilient infrastructure) as core strategic pillars.

5. The Sendai Framework's emphasis on "Build Back Better" reflects a conceptual shift from viewing disaster recovery as simple restoration to viewing it as an opportunity to reduce future vulnerability - for example, rebuilding damaged infrastructure to higher resilience standards rather than replicating pre-disaster vulnerabilities.

4. Institutions and policy tools

- FACT: **NDMA**: apex national disaster-management policy body, chaired by the Prime Minister.
- FACT: **National Institute of Disaster Management (NIDM)**: provides training, capacity-building and research support for disaster-management professionals.
- FACT: **National Disaster Response Force (NDRF)**: specialised, trained force for disaster response operations.
- FACT: **India Meteorological Department (IMD)**: provides early-warning data (cyclone, extreme-weather forecasting) central to disaster preparedness.

5. Indian applications and examples

- ANALYSIS: India's cyclone early-warning and evacuation systems (particularly on the east coast, e.g., Odisha's documented improvements since major historical cyclone events) illustrate substantial preparedness-capacity improvement over time, contributing to significantly reduced casualty figures in more recent major cyclone events compared to historical precedents.
- ANALYSIS: Post-disaster reconstruction efforts (e.g., following major flood or cyclone events) have increasingly incorporated resilience-focused rebuilding principles, reflecting the Sendai Framework's "Build Back Better" influence on Indian disaster-recovery practice.
- ANALYSIS: India's National Disaster Management Plan explicitly references alignment with the Sendai Framework's four priorities for action.

6. Must-Know Facts for Prelims

- FACT: The Disaster Management Act, 2005 established a three-tier structure: NDMA (chaired by the Prime Minister), SDMA (chaired by the Chief Minister) and DDMA (chaired by the District Magistrate/Collector).
- FACT: The Sendai Framework for Disaster Risk Reduction 2015-2030 was adopted in 2015 at the UN's Third World Conference on Disaster Risk Reduction in Sendai, Japan, succeeding the Hyogo Framework for Action.
- FACT: The Sendai Framework articulates four priorities for action: understanding disaster risk, strengthening disaster-risk governance, investing in disaster-risk reduction for resilience, and enhancing preparedness for effective response and "Build Back Better."
- FACT: Sendai also sets **seven global targets (A-G)** - substantially **reduce** (A) disaster mortality, (B) the number of affected people, (C) direct economic loss relative to global GDP, and (D) damage to critical infrastructure and disruption of basic services; and substantially **increase** (E) the number of countries with national and local DRR strategies, (F) international cooperation to developing countries, and (G) availability of and

access to multi-hazard early warning systems and disaster risk information. ANALYSIS: Four priorities **and** seven targets - a question asking for "the global targets of the Sendai Framework" is asking for the second set, not the first.

- FACT: The **World Conference on Disaster Risk Reduction** has been held three times, each hosted by **Japan: Yokohama (1994), Kobe (2005) and Sendai (2015)**. Yokohama produced the *Yokohama Strategy for a Safer World* (10 principles, arising from the mid-term review of the International Decade for Natural Disaster Reduction); Kobe produced the **Hyogo Framework for Action 2005-2015** with **five priorities** and launched an International Early Warning Programme; Sendai produced the current framework - the **first major agreement of the post-2015 development agenda**.

- FACT: India's **National Disaster Management Plan, 2016** was the country's first such plan. It is aligned with the Sendai Framework, the SDGs and the Paris Agreement, and organises action under **five thematic areas**: understanding risk; inter-agency coordination; investing in DRR through **structural** measures; investing in DRR through **non-structural** measures; and capacity development. ANALYSIS: A documented critique is that, unlike Sendai or the SDGs, the NDMP sets **no goals, targets, timeframes or funding projections** of its own.

- FACT: The **Prime Minister's Ten-Point Agenda on Disaster Risk Reduction** was enunciated at the **Asian Ministerial Conference on Disaster Risk Reduction (AMCDRR), New Delhi, 2016**; each point maps to one or more Sendai priorities for action.

- FACT: The disaster-management cycle comprises mitigation, preparedness, response and recovery as continuously linked phases, not a single reactive event.

- FACT: The National Disaster Response Force (NDRF) is India's specialised disaster-response operational force.

7. UPSC traps

- WRONG: Disaster management refers only to post-disaster relief and response. -> It encompasses the full cycle: mitigation, preparedness, response and recovery.

- WRONG: The Sendai Framework is a legally binding international treaty. -> It is a voluntary, non-binding UN framework guiding national disaster-risk-reduction policy.

- WRONG: NDMA is chaired by the Union Home Minister. -> It is chaired by the Prime Minister.

- WRONG: The Sendai Framework was the first global disaster-risk-reduction framework. -> It succeeded the earlier Hyogo Framework for Action (2005-2015), which itself followed the Yokohama Strategy (1994).

- WRONG: The Sendai Framework's "global targets" and its "priorities for action" are the same list. -> There are **four priorities for action** and **seven global targets (A-G)** - a distinction UPSC has tested directly.

- WRONG: India's National Disaster Management Plan sets its own quantified targets like Sendai does. -> The NDMP is aligned with Sendai and the SDGs but does **not** set its own goals, targets, timeframes or funding projections - a documented critique.

- WRONG: "Build Back Better" means restoring exactly pre-disaster conditions as quickly as possible. -> It specifically means rebuilding to reduce future vulnerability and improve resilience, not merely replicating the pre-disaster state.

8. CURRENT: Current anchor

- CURRENT: **Disaster Management (Amendment) Act, 2025 (Act 10 of 2025)** came into force on **9 April 2025**. It updates the statutory architecture, including disaster databases and provisions concerning national crisis/high- level bodies and urban authorities.

- FACT: Use `../../../../Disaster-Management/basic/02_Indian-Legal-and-Institutional-Architecture.md` for detail rather than duplicating it here. NIDM/official text.

- CURRENT: India's National Disaster Management Plan continues to be updated in alignment with the

Sendai Framework's 2015-2030 priorities; verify the latest NDMA plan revision and any updated early-warning-system capability (e.g., for cyclones, floods, or heatwaves) against the most recent NDMA/IMD publication before citing specific implementation details.

ANALYSIS: Interpretation caution: casualty-reduction figures for specific disaster events (e.g., comparing cyclone-related fatalities across different years/events) should be attributed to their specific source and event for precise, defensible citation.

9. PYQ application

- **FACT: 2024 GS-III direct PYQ (250 words):** "What is disaster resilience? How is it determined? Describe various elements of a resilience framework. Also mention the global targets of the Sendai Framework for Disaster Risk Reduction (2015-2030)." **ANALYSIS:** Three distinct demands. For "elements of a resilience framework", use the standard DRR fields of action: a **policy framework** backed by legal and institutional mechanisms; **risk assessment** based on hazard and community resilience; **risk awareness** among stakeholders and decision-makers; **implementation** through environmental management and urban planning; **early-warning systems**; and **use of knowledge** through informed stakeholder participation and accessible communication. Then list the **seven global targets (A-G)** - the question asks for targets, not the four priorities.

- **FACT: 2024 GS-III direct PYQ (250 words):** "Flooding in urban areas is an emerging climate-induced disaster. Discuss the causes of this disaster. Mention the features of two major floods in the last two decades in India. Describe the policies and frameworks aimed at dealing with such floods." **ANALYSIS:** The question demands **two named Indian urban floods with their features** - a factual requirement most answers skip. The documented challenges to cite are: inadequate comprehensive urban-flood risk assessment before mitigation planning; failure to map city-specific risk factors into development planning; weak inter-agency coordination; poor information sharing; disintegrated investment decisions; and lack of stakeholder consultation.

- **ANALYSIS:** Recurring Prelims pattern: correctly identify the NDMA-SDMA-DDMA chairing structure and the Sendai Framework's four priorities for action.

- **ANALYSIS:** Mains linkage: the "Build Back Better" principle is used to argue for resilience-focused (not merely restorative) post-disaster reconstruction policy.

10. Mains angles

- **ANALYSIS:** Argue that effective disaster governance requires treating mitigation and preparedness (proactive, risk-reduction-oriented) as equally important to response and recovery (reactive), reflecting the full disaster-management-cycle logic.

- **ANALYSIS:** Use the "Build Back Better" principle to argue that post-disaster reconstruction should be an opportunity for resilience investment, not simple restoration.

- **ANALYSIS:** Conclude with a risk-governance thesis: disaster resilience depends on institutional coordination (NDMA-SDMA-DDMA) working effectively at every level, especially the often under-resourced district level where actual implementation occurs.

Answer thesis: Treat disaster management as a continuous, four-phase cycle (not a single reactive response), and use the Sendai Framework's four priorities and "Build Back Better" principle as the analytical benchmark for evaluating India's disaster-risk-governance effectiveness at every institutional tier (NDMA-SDMA-DDMA).

11. Probable questions

- **ANALYSIS: Prelims:** Identify the correct chairing structure of NDMA, SDMA and DDMA, and the Sendai Framework's four priorities for action.

- **ANALYSIS: Mains (10 marks):** Explain the "Build Back Better" principle and its significance for India's post-disaster reconstruction policy.

- **ANALYSIS: Mains (15 marks):** Discuss the institutional strengths and implementation challenges of India's three-tier disaster-management structure under the Disaster Management Act, 2005.

12. Study links

- FACT: Advanced companion: advanced/26_Disaster-Management-Framework-and-Sendai.md.
- FACT: 17_Climate-Change-Science-Greenhouse-Effect.md - climate change's role in intensifying certain disaster-risk categories (extreme weather events).
- FACT: 24_Coastal-and-Marine-Ecology-CRZ-Blue-Economy.md - coastal-disaster-risk linkages relevant to cyclone preparedness.
- FACT: 27_Environmental-Institutions-MoEFCC-CPCB-NBA-WII.md - NDMA's institutional relationship with other environmental governance bodies.

13. Core answer architecture (10/15/20-mark support)

13.1 Direct demand A - resilience framework and Sendai targets (2024 GS-III, 15 marks)

Thesis: disaster resilience is the capacity of people, systems and infrastructure to anticipate, absorb, adapt and recover without recreating risk; it is determined by hazard exposure, vulnerability, coping capacity, institutions, information and the quality of recovery.

Answer spine: definition/measurement logic -> resilience elements (risk assessment and land-use; legal/institutional coordination; resilient infrastructure and finance; ecosystem buffers; early warning and inclusive participation; preparedness/Build Back Better) -> seven Sendai global targets **A-D reduce mortality, affected people, economic loss and critical-infrastructure/basic-service disruption; E-G increase national/local strategies, international cooperation and multi-hazard early-warning/risk-information access** -> DDMA/ULB implementation verdict. Do not substitute the four priorities for the seven targets.

13.2 Direct demand B - urban floods (2024 GS-III, 15 marks)

Demand part	Core material
Causes	Extreme rainfall/climate signal interacting with sealed surfaces, encroached wetlands/floodplains, inadequate or blocked drainage, construction/land-use decisions, upstream releases and tidal/backflow where relevant.
Two cases	Mumbai 2005: exceptionally intense rainfall met drainage, low-lying/reclaimed-land and waste-blockage vulnerabilities. Chennai 2015: northeast-monsoon extreme rainfall combined with altered wetlands/waterways and urban drainage/land-use stress. Use each as a causal feature, not a decorative name.
Frameworks	NDMA Guidelines on Management of Urban Flooding (2010) ; floodplain/zoning and drainage-catchment design; CWC-IMD warning/nowcasting; ULB/EOC response; Urban Flood Risk Management Programme; wetland restoration and risk-sensitive urban planning.

250-word spine: climate-risk framing -> physical and governance causes -> two compact cases -> structural/non-structural/institutional tools -> conclusion: restore storage and govern land use before relying on emergency pumping.

13.3 General 10/15/20 architecture

Use the Sendai shift from response to risk reduction, then map NDMA-SDMA-DDMA responsibilities, show a named ecosystem-based measure (mangrove/wetland/forest), identify the district/ULB capacity gap and end with a targeted, inclusive resilience verdict. Disaster-Management basic/08 supplies deeper flood evidence; this Core contains a complete answer route.

ENVIRONMENT AND ECOLOGY DEEP-REVIEW CORE CONTROL

- **Must remember:** Environment Topic 26 owns the climate-ecosystem-Sendai overlap: hazard, exposure, vulnerability and capacity produce risk, while the full disaster cycle and institutional architecture remain cross-owned by Disaster Management.
- **Close distinction:** Hazard is not disaster, resilience is not mere recovery, NDMA is not NEC, Sendai priorities are not its seven global targets, and global framework language is not a domestic statutory power.
- **Mechanism / status / evidence limit:** Preserve dedicated Disaster Management ownership; map prevention, mitigation, preparedness, response, recovery and build-back-better with exact national/state/district mandates and dated Sendai indicators.

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Hazard boundary?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: A. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q2. Which option preserves the ecological boundary of Hazard boundary?

A. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. B. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. C. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. D. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

Answer: B. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q3. Which statement uses Hazard boundary without changing its scale, parameter or status?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Answer: C. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q4. Which option avoids the standard UPSC close-option trap about Hazard boundary?

A. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. B. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. C. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. D. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.

Answer: D. Explanation: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q5. Which statement correctly identifies Exposure boundary?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. D. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

Answer: A. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q6. Which option preserves the ecological boundary of Exposure boundary?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Answer: B. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q7. Which statement uses Exposure boundary without changing its scale, parameter or status?

A. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. B. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. C. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: C. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q8. Which option avoids the standard UPSC close-option trap about Exposure boundary?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: D. Explanation: Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q9. Which statement correctly identifies Vulnerability boundary?

A. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Answer: A. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q10. Which option preserves the ecological boundary of Vulnerability boundary?

A. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: B. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q11. Which statement uses Vulnerability boundary without changing its scale, parameter or status?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: C. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q12. Which option avoids the standard UPSC close-option trap about Vulnerability boundary?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Answer: D. Explanation: Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q13. Which statement correctly identifies Capacity boundary?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: A. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q14. Which option preserves the ecological boundary of Capacity boundary?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: B. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q15. Which statement uses Capacity boundary without changing its scale, parameter or status?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. C. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: C. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q16. Which option avoids the standard UPSC close-option trap about Capacity boundary?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: D. Explanation: Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q17. Which statement correctly identifies Risk relation?

A. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. B. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. C. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: A. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q18. Which option preserves the ecological boundary of Risk relation?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: B. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q19. Which statement uses Risk relation without changing its scale, parameter or status?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

Answer: C. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q20. Which option avoids the standard UPSC close-option trap about Risk relation?

A. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. D. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

Answer: D. Explanation: Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q21. Which statement correctly identifies Disaster boundary?

A. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: A. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q22. Which option preserves the ecological boundary of Disaster boundary?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

Answer: B. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q23. Which statement uses Disaster boundary without changing its scale, parameter or status?

A. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

Answer: C. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q24. Which option avoids the standard UPSC close-option trap about Disaster boundary?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Answer: D. Explanation: A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q25. Which statement correctly identifies Prevention-mitigation distinction?

A. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. B. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

Answer: A. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q26. Which option preserves the ecological boundary of Prevention-mitigation distinction?

A. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. B. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

Answer: B. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q27. Which statement uses Prevention-mitigation distinction without changing its scale, parameter or status?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. C. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: C. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q28. Which option avoids the standard UPSC close-option trap about Prevention-mitigation distinction?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.

Answer: D. Explanation: Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q29. Which statement correctly identifies Preparedness boundary?

A. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

Answer: A. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q30. Which option preserves the ecological boundary of Preparedness boundary?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. C. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: B. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q31. Which statement uses Preparedness boundary without changing its scale, parameter or status?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: C. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q32. Which option avoids the standard UPSC close-option trap about Preparedness boundary?

A. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. B. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. C. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. D. Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.

Answer: D. Explanation: Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q33. Which statement correctly identifies Response boundary?

A. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: A. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q34. Which option preserves the ecological boundary of Response boundary?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: B. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q35. Which statement uses Response boundary without changing its scale, parameter or status?

A. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. B. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. C. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: C. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q36. Which option avoids the standard UPSC close-option trap about Response boundary?

A. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.

Answer: D. Explanation: Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q37. Which statement correctly identifies Recovery boundary?

A. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. B. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: A. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q38. Which option preserves the ecological boundary of Recovery boundary?

A. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. B. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: B. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q39. Which statement uses Recovery boundary without changing its scale, parameter or status?

A. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: C. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q40. Which option avoids the standard UPSC close-option trap about Recovery boundary?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. D. Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

Answer: D. Explanation: Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q41. Which statement correctly identifies Build-Back-Better boundary?

A. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: A. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q42. Which option preserves the ecological boundary of Build-Back-Better boundary?

A. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. B. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. C. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: B. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q43. Which statement uses Build-Back-Better boundary without changing its scale, parameter or status?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: C. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q44. Which option avoids the standard UPSC close-option trap about Build-Back-Better boundary?

A. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. B. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

Answer: D. Explanation: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q45. Which statement correctly identifies Cycle-continuum boundary?

A. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. B. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. C. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: A. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q46. Which option preserves the ecological boundary of Cycle-continuum boundary?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: B. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q47. Which statement uses Cycle-continuum boundary without changing its scale, parameter or status?

A. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. B. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. C. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. D. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Answer: C. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q48. Which option avoids the standard UPSC close-option trap about Cycle-continuum boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. D. Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Answer: D. Explanation: Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q49. Which statement correctly identifies Domestic-law boundary?

A. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: A. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q50. Which option preserves the ecological boundary of Domestic-law boundary?

A. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. B. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. C. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. D. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Answer: B. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q51. Which statement uses Domestic-law boundary without changing its scale, parameter or status?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: C. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q52. Which option avoids the standard UPSC close-option trap about Domestic-law boundary?

A. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. B. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. C. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. D. India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.

Answer: D. Explanation: India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q53. Which statement correctly identifies NDMA-NEC boundary?

A. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. D. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Answer: A. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q54. Which option preserves the ecological boundary of NDMA-NEC boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: B. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q55. Which statement uses NDMA-NEC boundary without changing its scale, parameter or status?

A. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. B. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. C. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: C. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q56. Which option avoids the standard UPSC close-option trap about NDMA-NEC boundary?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. D. NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.

Answer: D. Explanation: NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q57. Which statement correctly identifies NIDM-NDRF boundary?

A. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. B. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: A. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q58. Which option preserves the ecological boundary of NIDM-NDRF boundary?

A. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. B. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. C. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: B. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q59. Which statement uses NIDM-NDRF boundary without changing its scale, parameter or status?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: C. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q60. Which option avoids the standard UPSC close-option trap about NIDM-NDRF boundary?

A. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. B. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.

Answer: D. Explanation: NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q61. Which statement correctly identifies State-district boundary?

A. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: A. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q62. Which option preserves the ecological boundary of State-district boundary?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. C. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: B. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q63. Which statement uses State-district boundary without changing its scale, parameter or status?

A. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. B. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. C. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. D. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.

Answer: C. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q64. Which option avoids the standard UPSC close-option trap about State-district boundary?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. C. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. D. SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Answer: D. Explanation: SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q65. Which statement correctly identifies Sendai identity?

A. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. B. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. C. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: A. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q66. Which option preserves the ecological boundary of Sendai identity?

A. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. B. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. C. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. D. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.

Answer: B. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q67. Which statement uses Sendai identity without changing its scale, parameter or status?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. C. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: C. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q68. Which option avoids the standard UPSC close-option trap about Sendai identity?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.

Answer: D. Explanation: The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q69. Which statement correctly identifies Priorities-targets boundary?

A. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. D. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.

Answer: A. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q70. Which option preserves the ecological boundary of Priorities-targets boundary?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: B. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q71. Which statement uses Priorities-targets boundary without changing its scale, parameter or status?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. C. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. D. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Answer: C. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q72. Which option avoids the standard UPSC close-option trap about Priorities-targets boundary?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. D. Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.

Answer: D. Explanation: Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q73. Which statement correctly identifies Global-domestic boundary?

A. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. B. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. C. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. D. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Answer: A. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q74. Which option preserves the ecological boundary of Global-domestic boundary?

A. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. B. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. C. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. D. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Answer: B. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q75. Which statement uses Global-domestic boundary without changing its scale, parameter or status?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. C. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: C. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q76. Which option avoids the standard UPSC close-option trap about Global-domestic boundary?

A. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. D. Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Answer: D. Explanation: Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q77. Which statement correctly identifies Current evidence boundary?

A. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. B. A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. C. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. D. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Answer: A. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q78. Which option preserves the ecological boundary of Current evidence boundary?

A. Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. B. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. C. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. D. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Answer: B. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q79. Which statement uses Current evidence boundary without changing its scale, parameter or status?

A. Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. B. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. C. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. D. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.

Answer: C. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

Q80. Which option avoids the standard UPSC close-option trap about Current evidence boundary?

A. Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. B. Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. C. A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. D. Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Answer: D. Explanation: Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The other options belong to different media, parameters, processes, scales, actors, institutions, instruments or status categories.

PYQS AND ANSWER PRACTICE

AUDITED DISASTER RISK, RESILIENCE, SENDAI AND URBAN-FLOOD PYQ OWNERSHIP

Audited ledgers route the verified 2024 resilience, Sendai-target and urban-flood demands plus related disaster-management questions. No objective key, loss figure, target progress or official model answer is inferred.

Demand decoding: The directive **answer** requires a direct position on "AUDITED DISASTER RISK, RESILIENCE, SENDAI AND URBAN-FLOOD PYQ OWNERSHIP", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: The answer must resolve the Environment and Ecology demand in "AUDITED DISASTER RISK, RESILIENCE, SENDAI AND URBAN-FLOOD PYQ OWNERSHIP".

Analytical body:

1. **Claim and named evidence:** AUDITED DISASTER RISK, RESILIENCE, SENDAI AND URBAN-FLOOD PYQ OWNERSHIP **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Audited ledgers route the verified 2024 resilience, Sendai-target and urban-flood demands plus related disaster-management questions. No objective key, loss figure, target progress or official model answer is inferred. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: The answer must resolve the Environment and Ecology demand in "AUDITED DISASTER RISK, RESILIENCE, SENDAI AND URBAN-FLOOD PYQ OWNERSHIP".

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "AUDITED DISASTER RISK, RESILIENCE, SENDAI AND URBAN-FLOOD PYQ OWNERSHIP", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

OWNER PYQ LEDGER EXTRACTS

9. PYQ application

- **FACT: 2024 GS-III direct PYQ (250 words):** "What is disaster resilience? How is it determined? Describe various elements of a resilience framework. Also mention the global targets of the Sendai Framework for Disaster Risk Reduction (2015-2030)." **ANALYSIS:** Three distinct demands. For "elements of a resilience framework", use the standard DRR fields of action: a **policy framework** backed by legal and institutional mechanisms; **risk assessment** based on hazard and community resilience; **risk awareness** among stakeholders and decision-makers; **implementation** through environmental management and urban planning; **early-warning systems**; and **use of knowledge** through informed stakeholder participation and accessible communication. Then list the

seven global targets (A-G) - the question asks for targets, not the four priorities.

- **FACT: 2024 GS-III direct PYQ (250 words):** "Flooding in urban areas is an emerging climate-induced disaster. Discuss the causes of this disaster. Mention the features of two major floods in the last two decades in India. Describe the policies and frameworks aimed at dealing with such floods." **ANALYSIS:** The question demands **two named Indian urban floods with their features** - a factual requirement most answers skip. The documented challenges to cite are: inadequate comprehensive urban-flood risk assessment before mitigation planning; failure to map city-specific risk factors into development planning; weak inter-agency coordination; poor information sharing; disintegrated investment decisions; and lack of stakeholder consultation.
- **ANALYSIS:** Recurring Prelims pattern: correctly identify the NDMA-SDMA-DDMA chairing structure and the Sendai Framework's four priorities for action.
- **ANALYSIS:** Mains linkage: the "Build Back Better" principle is used to argue for resilience-focused (not merely restorative) post-disaster reconstruction policy.

10. PYQ-based analytical application

- **ANALYSIS:** Prelims questions on the Sendai Framework's priorities or India's three-tier structure should be answered by applying the precise priority-order and chairing-structure facts.
- **ANALYSIS:** Mains answers on "strengthening disaster management in India" should explicitly engage the DDMA-capacity-gap critique and the ecosystem-based-risk-reduction integration point to demonstrate analytical depth beyond describing the institutional structure.

PYQ DEMAND CARD 1 - 2024 GS-III

Demand: Define disaster resilience, explain its determination and elements, and mention Sendai global targets.

Status: Verified direct Mains demand; four priorities are not substituted for seven targets.

Model solution: Hazard boundary: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Build-Back-Better boundary:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Priorities-targets boundary:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Global-domestic boundary:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 1 - 2024 GS-III", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Hazard boundary: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Build-Back-Better**

boundary: Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Priorities-targets boundary:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Global-domestic boundary:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Analytical body:

1. **Claim and named evidence:** Demand: Define disaster resilience, explain its determination and elements, and mention Sendai global targets. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Status: Verified direct Mains demand; four priorities are not substituted for seven targets. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: **Hazard boundary:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Build-Back-Better boundary:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Priorities-targets boundary:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Global-domestic boundary:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "PYQ DEMAND CARD 1 - 2024 GS-III", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

PYQ DEMAND CARD 2 - 2024 GS-III

Demand: Discuss causes, cases and policy frameworks for urban flooding in India.

Status: Verified direct Mains demand; no disaster-loss value is inferred.

Model solution: Hazard boundary: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Prevention-mitigation distinction:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Preparedness boundary:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Domestic-law boundary:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **State-district boundary:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Current evidence boundary:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Demand decoding: The directive **answer** requires a direct position on "PYQ DEMAND CARD 2 - 2024 GS-III", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Hazard boundary: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Prevention-mitigation distinction:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Preparedness boundary:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Domestic-law boundary:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **State-district boundary:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Current evidence boundary:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Analytical body:

- 1. Claim and named evidence:** Demand: Discuss causes, cases and policy frameworks for urban flooding in India. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
- 2. Claim and named evidence:** Status: Verified direct Mains demand; no disaster-loss value is inferred. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow,

parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Hazard boundary: A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Prevention-mitigation distinction:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Preparedness boundary:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Domestic-law boundary:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **State-district boundary:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Current evidence boundary:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. The answer therefore separates definition, mechanism, evidence and limitation, and does not infer an official model answer or objective key.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "PYQ DEMAND CARD 2 - 2024 GS-III", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish hazard, exposure, vulnerability, capacity, risk and disaster. Answer in about 150 words.

Model thesis: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources,

institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Disaster boundary. **Named evidence/example:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
- Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.
- Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.
- Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.
- Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.
- A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.

Qualified conclusion: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible

actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Disaster boundary. **Named evidence/example:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **answer** requires a direct position on "Distinguish hazard, exposure, vulnerability, capacity, risk and disaster. Answer in about 150...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Disaster boundary. **Named evidence/example:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Analysis:** Connect the defined

system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

3. **Claim and named evidence:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
5. **Claim and named evidence:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
6. **Claim and named evidence:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Disaster boundary. **Named evidence/example:** A disaster is serious disruption

and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Distinguish hazard, exposure, vulnerability, capacity, risk and disaster. Answer in about 150...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 2 - 10 MARKS

Question: Explain prevention, mitigation and preparedness as distinct phases. Answer in about 150 words.

Model thesis: Claim: Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.
- Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.
- Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Qualified conclusion: Claim: Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must

retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **explain** requires a direct position on "Explain prevention, mitigation and preparedness as distinct phases. Answer in about 150 words.", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: **Claim:** Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather

than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 10 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Explain prevention, mitigation and preparedness as distinct phases. Answer in about 150 words.", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain response, recovery and Build Back Better. Answer in about 250 words.

Model thesis: **Claim:** Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.
- Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

- Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.
- Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.

Qualified conclusion: Claim: Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **explain** requires a direct position on "Explain response, recovery and Build Back Better. Answer in about 250 words.", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: Claim: Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: **Claim:** Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Cycle-continuum boundary. **Named evidence/example:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Explain response, recovery and Build Back Better. Answer in about 250 words.", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 4 - 15 MARKS

Question: Map NDMA, NEC, NIDM, NDRF, SDMA and DDMA by role. Answer in about 250 words.

Model thesis: **Claim:** Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NIDM-NDRF boundary. **Named evidence/example:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
- NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.
- NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.
- SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.

Qualified conclusion: **Claim:** Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NIDM-NDRF boundary. **Named evidence/example:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage,

legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **answer** requires a direct position on "Map NDMA, NEC, NIDM, NDRF, SDMA and DDMA by role. Answer in about 250 words.", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NIDM-NDRF boundary. **Named evidence/example:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit,

source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NIDM-NDRF boundary. **Named evidence/example:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 15 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Map NDMA, NEC, NIDM, NDRF, SDMA and DDMA by role. Answer in about 250 words.", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 5 - 20 MARKS

Question: Compare Sendai's global framework with India's domestic legal architecture. Answer in about 300 words.

Model thesis: Claim: Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and

implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sendai identity. **Named evidence/example:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Priorities-targets boundary. **Named evidence/example:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Global-domestic boundary. **Named evidence/example:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
- NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.
- SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.
- The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.
- Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.
- Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.

Qualified conclusion: **Claim:** Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sendai identity. **Named evidence/example:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. **Analysis:** This identifies

the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Priorities-targets boundary. **Named evidence/example:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Global-domestic boundary. **Named evidence/example:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Demand decoding: The directive **compare** requires a direct position on "Compare Sendai's global framework with India's domestic legal architecture. Answer in about...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sendai identity. **Named evidence/example:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Priorities-targets boundary. **Named evidence/example:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Global-domestic boundary. **Named evidence/example:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Analytical body:

1. **Claim and named evidence:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
3. **Claim and named evidence:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
4. **Claim and named evidence:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
5. **Claim and named evidence:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
6. **Claim and named evidence:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.

Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Domestic-law boundary. **Named evidence/example:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** NDMA-NEC boundary. **Named evidence/example:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** State-district boundary. **Named evidence/example:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Sendai identity. **Named evidence/example:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make

the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Priorities-targets boundary. **Named evidence/example:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Global-domestic boundary. **Named evidence/example:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Compare Sendai's global framework with India's domestic legal architecture. Answer in about...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design a risk-reduction framework using the full disaster-management continuum. Answer in about 300 words.

Model thesis: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Risk relation. **Named evidence/example:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's

ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Build-Back-Better boundary. **Named evidence/example:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Current evidence boundary. **Named evidence/example:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim -> named evidence -> analysis -> qualification:

- A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
- Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.
- Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.
- Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.
- Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.
- Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.
- Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.
- Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.
- Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.

- Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.
- Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Qualified conclusion: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; 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Demand decoding: The directive **answer** requires a direct position on "Design a risk-reduction framework using the full disaster-management continuum. Answer in...", every clause, exact ecological mechanism and scale, named Indian law or institution, dated treaty/policy status, evidence, trade-offs and a qualified conclusion.

Detailed examiner-grade model answer:

Introduction and thesis: **Claim:** Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Vulnerability boundary. **Named evidence/example:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.

Analysis: This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Capacity boundary. **Named evidence/example:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.

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Claim: Prevention-mitigation distinction. **Named evidence/example:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Preparedness boundary. **Named evidence/example:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner.

Claim: Response boundary. **Named evidence/example:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Recovery boundary. **Named evidence/example:** Recovery restores and

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Analytical body:

1. **Claim and named evidence:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
2. **Claim and named evidence:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
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6. **Claim and named evidence:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented. **Analysis:** Connect the defined system/taxon and named evidence -> ecological mechanism or legal/institutional instrument -> implementation pathway -> ecological and social consequence. **Qualification:** State scale, stock/flow, parameter/unit, source/date/status, jurisdiction, uncertainty, causal limit, exception or residual risk.
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Counter-position / limit: A designation, schedule, COP decision, policy target, budget, installed capacity, registration, treatment capacity or chronological association cannot alone establish ecological recovery, compliance, attribution or net climate benefit; test mechanism, monitoring, counterfactual and implementation.

Qualified conclusion: Claim: Hazard boundary. **Named evidence/example:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk. **Analysis:** This identifies the environmental mechanism, system boundary, measured parameter, responsible actor and causal link that make the claim examinable. **Qualification:** The conclusion must retain the source's ecological or regulatory scale, temporal stage, rule vintage, legal status, metric and evidence boundary rather than generalise beyond the owner. **Claim:** Exposure boundary. **Named evidence/example:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.

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Executable exam-length answer / compression plan: For 20 marks, spend one-sixth of the time decoding the directive and drawing definition -> mechanism/status -> institution/instrument -> implementation -> ecological and social outcome; write four to seven claim -> named evidence -> analysis -> qualification points; reserve the final minute for taxon, unit, baseline, date, jurisdiction, exception, causation and residual-risk checks.

Why this earns marks: The answer obeys the directive, explains mechanisms rather than listing schemes, uses India-centric evidence and preserves ecological, species, legal, treaty, institutional, climate-unit, pollution-standard and causal distinctions.

How to improve this answer: For "Design a risk-reduction framework using the full disaster-management continuum. Answer in...", replace the weakest catalogue point with one exact mechanism or distinction, named law/institution/treaty/species, dated status, measurable outcome, implementation constraint and answer-specific qualification.

OPTIONAL ADVANCED DEPTH - NOT REQUIRED FOR A CORE ANSWER

Subject: Environment and Ecology | **Tier:** Advanced | **GS Paper:** GS-III (Disaster Management) + Prelims. **Core area:** Disaster-risk-reduction governance architecture. **Grounded in:** Disaster Management Act, 2005 full text; Sendai Framework for Disaster Risk Reduction 2015-2030 (UNDRR) monitoring indicators; National Disaster Management Plan revisions; audited UPSC Environment/Disaster Management PYQs (2024-2026 Prelims and 2024-2025 Mains). **FACT:** = source-grounded | **ANALYSIS:** = inference/analysis | **CURRENT:** = dated current-affairs anchor. **Companion:** basic/26_Disaster-Management-Framework-and-Sendai.md.

1. The paradigm shift Sendai represents: from "disaster management" to "disaster risk reduction"

ANALYSIS: The single most important advanced-level conceptual point: the global disaster-policy field underwent a genuine paradigm shift - from the Yokohama Strategy (1994) and Hyogo Framework (2005-2015) era's focus on *managing* disasters as they occur, toward the Sendai Framework's (2015-2030) explicit focus on *reducing existing disaster risk* and *preventing the creation of new risk* through development planning itself (e.g., land-use planning, building codes, ecosystem-based approaches) - treating disaster risk as a systemic development-planning issue rather than solely an emergency-response issue. This reframing is why Sendai's four priorities emphasise "understanding risk" and "risk governance" ahead of "preparedness for effective response" - response capability remains essential, but is positioned as the framework's fourth priority, not its starting point.

2. India's DDMA-level implementation gap: the recurring, examinable weak link

1. FACT: The Disaster Management Act, 2005's three-tier structure places DDMA (District Disaster Management Authority) as the primary implementing body closest to actual disaster events, chaired by the District Magistrate/Collector.
2. ANALYSIS: **Documented implementation critique:** district-level disaster-management capacity (dedicated staffing, technical expertise, updated hazard/vulnerability mapping, and resourcing for mitigation infrastructure) has been repeatedly flagged by various parliamentary and expert reviews as the weakest link in India's disaster-governance chain - District Magistrates typically hold disaster-management responsibilities alongside numerous other administrative duties, without always having a dedicated, full-time, technically specialised disaster-management team comparable to state or national-level bodies.
3. ANALYSIS: **Analytical synthesis:** this creates a structural mismatch - the institutional tier with the most detailed local knowledge and the fastest initial-response role (DDMA) is often the least resourced tier in practice, while national/state bodies (NDMA/SDMA) with greater resources and policy authority are geographically and administratively more distant from the actual disaster event - a recurring theme when Mains questions ask about strengthening India's disaster-governance architecture.

3. Ecosystem-based disaster-risk reduction: a cross-topic integration point

- ANALYSIS: The Sendai Framework and India's evolving disaster policy increasingly recognise ecosystem-based approaches - healthy mangroves buffering cyclone storm surge (Topic 24), intact wetlands absorbing flood peaks (Topic 07), and stable forest cover reducing landslide/erosion risk (Topics 03, 11-12) - as disaster-risk-reduction infrastructure in their own right, not merely as separate "environmental" concerns.
- ANALYSIS: **Analytical significance:** this framing directly integrates disaster-management policy with the ecosystem-services concept introduced in Topic 01 - degrading a mangrove forest for short-term coastal development is not just a biodiversity loss but a disaster-risk-increasing decision, since it removes a "regulating service" (storm-surge buffering) that has quantifiable value in disaster-risk terms. A sophisticated Mains answer on disaster management should explicitly cite ecosystem-based measures alongside conventional "grey" (engineered) infrastructure as part of a comprehensive risk-reduction strategy.

4. Early-warning systems: documented success and continuing gaps

- FACT: India's cyclone early-warning and evacuation capability (led by IMD forecasting combined with NDMA/state-government coordinated evacuation protocols) has been credited with substantially reducing cyclone-related casualties in more recent major events compared to historical precedents from decades earlier - a genuine institutional success story worth citing with specific event comparisons where reliably documented.
- ANALYSIS: **Continuing gaps:** early-warning and preparedness capability is markedly stronger for well-studied, more predictable hazards (cyclones, seasonal floods) than for less predictable or more localised hazards (flash floods, urban flooding, landslides, and heatwave-specific early-action protocols), which have received comparatively newer and less mature institutional attention - heatwave action plans, for instance, are a more recently developed instrument compared to the decades-long evolution of cyclone- preparedness systems.

5. Data and conceptual limitations

- ANALYSIS: Comparative casualty-reduction claims across different disaster events (e.g., citing

that a specific recent cyclone caused markedly fewer deaths than a historical comparator) should always cite the specific events and official casualty figures being compared, since disaster-mortality data can vary in reporting completeness and definition across different events and time periods.

- ANALYSIS: District-level disaster-management capacity assessments (staffing, resourcing, technical capability) are not comprehensively, uniformly published in a single national dataset; specific capacity claims should be attributed to the relevant parliamentary committee report, NDMA assessment, or academic study cited.
- ANALYSIS: Quantifying the specific risk-reduction value of ecosystem-based measures (e.g., exact storm-surge-height reduction attributable to a specific mangrove belt) requires site-specific scientific studies rather than a single universal figure.

6. Recurring UPSC analytical tensions

Tension	Balanced framing
National/state disaster-policy authority vs district-level implementation capacity	Genuine disaster resilience requires resourcing the DDMA tier commensurate with its critical, closest-to-ground implementation role, not just strengthening national/state policy frameworks.
Engineered ("grey") disaster infrastructure vs ecosystem-based ("green") risk reduction	Both are valuable and increasingly viewed as complementary; ecosystem-based measures (mangroves, wetlands, forests) should be integrated into disaster-risk-reduction planning, not treated as a separate environmental-policy silo.
Mature hazard-specific preparedness (cyclones) vs newer, less-developed hazard preparedness (heatwaves, urban flooding)	India's disaster-preparedness capability is uneven across hazard types; newer instruments (heatwave action plans) require continued institutional development comparable to the decades-long maturation of cyclone-preparedness systems.

7. Must-Know Facts for Advanced Prelims

- FACT: The Sendai Framework represents a paradigm shift from "disaster management" (managing events as they occur) to "disaster risk reduction" (preventing/reducing risk through development planning itself), succeeding the Hyogo Framework for Action (2005-2015).
- FACT: District Disaster Management Authorities (DDMAs), chaired by the District Magistrate/Collector, have been repeatedly flagged in reviews as the least-resourced tier in India's three-tier disaster-governance structure despite their critical, closest-to-ground implementation role.
- FACT: Ecosystem-based disaster-risk reduction (mangroves, wetlands, forests as risk-reduction infrastructure) is an increasingly recognised complement to engineered disaster infrastructure under both Sendai Framework principles and evolving Indian policy.
- FACT: India's cyclone early-warning/evacuation capability has matured substantially further than its preparedness capability for hazards like heatwaves or urban flooding, which are comparatively newer areas of institutional focus.

8. Advanced Prelims traps

- WRONG: The Sendai Framework primarily emphasises improving emergency response capability as its first priority. -> Its first priority is "understanding disaster risk," reflecting the paradigm shift toward risk reduction and prevention rather than response-first framing.
- WRONG: India's three-tier disaster-management structure is uniformly well-resourced across all levels. -> DDMA (district level) has been repeatedly flagged as comparatively under- resourced relative to its critical implementation role.
- WRONG: Ecosystem-based disaster-risk reduction (mangroves, wetlands) is considered separate from and unrelated to formal disaster-management policy. -> It is increasingly integrated as a recognised risk-reduction approach under Sendai Framework principles and evolving Indian policy.
- WRONG: India's disaster-preparedness capability is uniformly mature across all hazard types. ->

It is notably stronger for well-studied hazards like cyclones than for newer areas like heatwave action planning or urban-flood preparedness.

9. CURRENT: Current anchor - analytical use

CURRENT: **Disaster Management (Amendment) Act, 2025**, in force 9 April 2025, is the current legal update. Route institutional detail to

.../Disaster-Management/advanced/02_Indian-Legal-and-Institutional-Architecture.md instead of duplicating it. NIDM/official text.

Verified current anchor	Topic-specific analytical use
CURRENT: India's National Disaster Management Plan, aligned with Sendai Framework priorities, continues to be updated (verify the latest NDMA plan revision and hazard-specific action-plan development, e.g., heatwave action plans, before citing specific current details).	Use as the current institutional baseline while pairing it with the DDMA-capacity-gap critique and the ecosystem-based-risk-reduction integration point for full analytical depth.
FACT: The Hyogo-to-Sendai shift is the analytical core. Hyogo (2005-2015) set five priorities framed around <i>managing disasters</i> ; Sendai (2015-2030) set four priorities plus seven measurable global targets framed around <i>managing risk</i> , and extended explicitly to "Build Back Better" in recovery, rehabilitation and reconstruction.	The examinable proposition: Sendai's decisive innovation was not new priorities but measurable targets with a baseline period , converting DRR from exhortation into something auditable. India's NDMP inherits the priorities but not the targets - which is precisely where the accountability gap sits.
FACT: Heatwave governance is the fastest-moving gap: heatwaves are currently treated largely as local disasters, while their intensifying severity argues for a broader national approach; documented weaknesses include the absence of sub-district-level data to build separate urban and rural heat indices, thin intra-city ward-level vulnerability analysis, and the lack of a settled financing arrangement for Heat Action Plans.	The single best contemporary illustration of the hazard-classification-drives-financing problem: how a hazard is legally and administratively categorised determines which fund pays for it. Use alongside Topic 17's SLCF/urban-heat science for a science-to-governance chain.
FACT: Ecosystem-based disaster risk reduction is now explicit in Indian fiscal policy - MISHTI (mangroves as surge buffers), NPCA-expanded wetlands as flood-moderation buffers, and the National Coastal Mission (Economic Survey 2025-26, Ch. 10).	Lets a disaster-management answer draw directly on Topics 01, 07 and 24: the cheapest disaster infrastructure is often an intact ecosystem, and the Government now says so in its own budget documents.

10. PYQ-based analytical application

- ANALYSIS: Prelims questions on the Sendai Framework's priorities or India's three-tier structure should be answered by applying the precise priority-order and chairing-structure facts.
- ANALYSIS: Mains answers on "strengthening disaster management in India" should explicitly engage the DDMA-capacity-gap critique and the ecosystem-based-risk-reduction integration point to demonstrate analytical depth beyond describing the institutional structure.

11. Mains-ready framework

Central thesis: India's disaster-governance architecture reflects the Sendai Framework's paradigm shift toward risk reduction and prevention, but its real effectiveness is constrained by a persistent capacity gap at the district level - the tier closest to actual disaster events - and by uneven institutional maturity across hazard types; genuine resilience requires resourcing DDMA's commensurately, integrating ecosystem-based risk-reduction measures alongside engineered infrastructure, and extending the same institutional maturity seen in cyclone preparedness to newer hazard categories like heatwaves and urban flooding.

1. Explain the Sendai Framework's paradigm shift from disaster management to disaster risk reduction, and its four priorities.
2. Present India's three-tier structure (NDMA-SDMA-DDMA) and the documented DDMA-capacity gap as the central implementation critique.
3. Integrate ecosystem-based risk-reduction measures (mangroves, wetlands, forests) as a cross-topic analytical point connecting disaster policy to ecosystem-services concepts.

4. Compare mature (cyclone) versus newer (heatwave, urban flooding) hazard-preparedness institutional development.
5. Conclude with a DDMA-resourcing and ecosystem-integration recommendation as the genuine determinants of improved disaster resilience.

12. Probable questions

- ANALYSIS: **Prelims:** Identify the Sendai Framework's four priorities for action in correct order and India's three-tier disaster-management chairing structure.
- ANALYSIS: **Mains (10 marks):** Why has the District Disaster Management Authority been identified as the weakest link in India's disaster-governance structure?
- ANALYSIS: **Mains (15 marks):** Discuss the role of ecosystem-based approaches (mangroves, wetlands, forests) in disaster-risk reduction, integrating this with India's formal disaster-management framework.

13. Study links

- FACT: Foundation companion: basic/26_Disaster-Management-Framework-and-Sendai.md.
- FACT: 24_Coastal-and-Marine-Ecology-CRZ-Blue-Economy.md - mangrove-based storm-surge buffering as ecosystem-based disaster-risk reduction.
- FACT: 07_Biosphere-Reserves-and-Ramsar-Sites.md - wetlands' flood-buffering ecosystem service.
- FACT: 01_Ecosystem-Structure-and-Function.md - the ecosystem-services framework underlying this topic's ecosystem-based risk-reduction integration.

CONSOLIDATED REGISTER NOTES

Disaster Management Framework and Sendai: HAZARD, EXPOSURE, VULNERABILITY, CAPACITY AND CYCLE MAP

1. **Hazard boundary:** A hazard is a potentially damaging process, phenomenon or human activity; it is not the same as a disaster or the complete measure of risk.
2. **Exposure boundary:** Exposure identifies people, infrastructure, livelihoods, ecosystems and assets located where hazards may occur; it does not describe their susceptibility.
3. **Vulnerability boundary:** Vulnerability is the set of physical, social, economic and environmental conditions that increase susceptibility to harm; it is distinct from exposure.
4. **Capacity boundary:** Capacity comprises strengths, resources, institutions and abilities available to manage and reduce risk; low capacity can magnify vulnerability and losses.
5. **Risk relation:** Disaster risk emerges from the interaction of hazard, exposure, vulnerability and capacity across a stated place and time; it cannot be inferred from hazard intensity alone.
6. **Disaster boundary:** A disaster is serious disruption and loss that exceeds or severely tests the affected community's ability to cope; a hazard event need not become a disaster.
7. **Prevention-mitigation distinction:** Prevention seeks to avoid existing and new disaster risk where possible, while mitigation reduces adverse impacts where risk cannot be fully prevented.
8. **Preparedness boundary:** Preparedness develops knowledge, plans, warnings, training, stocks and coordination for effective action before an event; it is not post-event response.
9. **Response boundary:** Response covers immediate and short-term actions to save lives, protect health and meet basic needs during or just after a disaster.
10. **Recovery boundary:** Recovery restores and improves livelihoods, systems and assets after emergency response, while rehabilitation and reconstruction are related but distinct components.
11. **Build-Back-Better boundary:** Build Back Better uses recovery, rehabilitation and reconstruction to reduce future risk rather than merely recreate pre-disaster vulnerability.

12. **Cycle-continuum boundary:** Prevention, mitigation, preparedness, response and recovery are linked phases with feedback; disaster management is not a linear one-time sequence.
13. **Domestic-law boundary:** India's Disaster Management Act creates domestic legal and institutional duties; it must not be treated as the legal implementation text of the Sendai Framework.
14. **NDMA-NEC boundary:** NDMA is the national authority for policy and guidelines, while the National Executive Committee supports coordination, implementation and national planning under the domestic framework.
15. **NIDM-NDRF boundary:** NIDM focuses on training, research and capacity development, while NDRF is a specialised response force; neither should be substituted for the other.
16. **State-district boundary:** SDMA and DDMA operate at state and district levels within domestic law; their plans and implementation roles must be attributed to the applicable statutory text.
17. **Sendai identity:** The Sendai Framework 2015-2030 is a global, voluntary and non-binding disaster-risk-reduction framework, not a domestic statute or operational response agency.
18. **Priorities-targets boundary:** Sendai has four priorities for action and seven global targets; the priorities describe fields of action, while the targets describe global reduction or increase directions.
19. **Global-domestic boundary:** Sendai's global targets guide international monitoring, while Indian institutions, plans, funds and legal powers arise from domestic law and policy; alignment does not erase this distinction.
20. **Current evidence boundary:** Disaster losses, mortality, affected people, target progress, indicators, warning coverage, institutional powers and plan status require dated official NDMA, India Code, UNDRR or sector-agency evidence.

Disaster Management Framework and Sendai: PHASE, INSTITUTION, PRIORITY, TARGET AND JURISDICTION TRAPS

- Do not merge hazard with disaster.
- Do not merge exposure with vulnerability.
- Do not omit capacity from a risk analysis.
- Do not infer risk from hazard intensity alone.
- Do not call every hazard event a disaster.
- Do not merge prevention with mitigation.
- Do not call preparedness a response activity.
- Do not call emergency response long-term recovery.
- Do not reduce recovery to rebuilding identical assets.
- Do not make the disaster cycle a one-way sequence.
- Do not derive domestic legal powers from Sendai.
- Do not merge NDMA with NEC.
- Do not merge NIDM training with NDRF operations.
- Do not move national roles automatically to a DDMA.
- Do not call Sendai a binding treaty.
- Do not replace four priorities with seven targets.
- Do not replace seven global targets with four priorities.
- Do not equate global monitoring with domestic enforcement.
- Do not invent losses, indicators or institutional powers.
- Do not state a current plan or amendment status without dated official text.

Disaster Management Framework and Sendai: DISASTER-RISK-REDUCTION ANSWER SPINE

DEFINE HAZARD EXPOSURE VULNERABILITY CAPACITY RISK AND DISASTER
-> SEPARATE PREVENTION MITIGATION PREPAREDNESS RESPONSE AND RECOVERY
-> USE BUILD BACK BETTER TO REDUCE FUTURE RISK
-> MAP NDMA NEC NIDM NDRF SDMA AND DDMA ONLY BY OWNED ROLE
-> SEPARATE SENDAI FOUR PRIORITIES FROM SEVEN GLOBAL TARGETS
-> SEPARATE GLOBAL FRAMEWORK FROM DOMESTIC LAW INSTITUTIONS AND FUNDS
-> CONCLUDE WITH DATED LAW PLAN LOSS INDICATOR AND WARNING EVIDENCE

Disaster Management Framework and Sendai: LIVE LAW, PLAN, LOSS, INDICATOR, WARNING AND TARGET EVIDENCE BOUNDARY

UNDRR supplied substantive Sendai framework text. Indian official routes did not yield substantive current legal or plan text in this check, so no disaster loss, target progress, indicator, warning coverage, amended power or plan-status claim is asserted.

COMPLETE TOPIC ASCII MASTER FLOW DIAGRAM

ASCII MASTER FLOW - PANEL 1/12: Disaster-risk equation

HAZARD -> potentially damaging process or event
EXPOSURE -> people assets systems in harm's way
VULNERABILITY -> susceptibility to damage
CAPACITY -> resources and ability to manage risk
RISK -> interaction across a stated place and time
MUST REMEMBER: Environment Topic 26 owns the climate-ecosystem-Sendai overlap: hazard,...
MUST REMEMBER: Environment Topic 26 owns the climate-ecosystem-Sendai overlap: hazard,...

ASCII MASTER FLOW - PANEL 2/12: Hazard-to-disaster threshold

HAZARD OCCURS -> physical or human-induced event
EXPOSED SYSTEM -> contact with people assets or ecosystems
VULNERABILITY AND LOW CAPACITY -> losses amplify
COPING ABILITY SEVERELY TESTED -> serious disruption
DISASTER -> not every hazard crosses this threshold

ASCII MASTER FLOW - PANEL 3/12: Risk-reduction ladder

PREVENTION -> avoid creation or existence of risk where possible
MITIGATION -> reduce adverse impact of remaining risk
PREPAREDNESS -> ready plans warnings skills and resources
RESPONSE -> immediate life-saving and basic needs
RECOVERY -> restore and improve while reducing future risk

ASCII MASTER FLOW - PANEL 4/12: Build Back Better loop

DAMAGE AND NEEDS ASSESSMENT -> establish recovery baseline
REHABILITATION -> restore essential functions
RECONSTRUCTION -> rebuild assets and systems
RISK-INFORMED IMPROVEMENT -> avoid old vulnerability
FEEDBACK -> prevention mitigation and preparedness improve

ASCII MASTER FLOW - PANEL 5/12: Domestic architecture firewall

DISASTER MANAGEMENT ACT -> domestic legal base
NDMA -> national policy and guidelines
NEC -> executive coordination and implementation support
SDMA AND DDMA -> state and district statutory levels
RULE -> roles come from domestic law, not Sendai text

ASCII MASTER FLOW - PANEL 6/12: Training-response split

NIDM -> training research documentation capacity development
NDRF -> specialised disaster response operations
NDMA -> policy and guidance
DDMA -> district planning and implementation context
RULE -> institution type determines the answer

ASCII MASTER FLOW - PANEL 7/12: Sendai identity card

PERIOD -> 2015 to 2030
NATURE -> global voluntary non-binding framework
PURPOSE -> prevent new and reduce existing disaster risk
SCOPE -> people livelihoods health assets and systems
NOT -> Indian statute fund or response force

ASCII MASTER FLOW - PANEL 8/12: Four priorities rail

PRIORITY 1 -> understand disaster risk
PRIORITY 2 -> strengthen risk governance
PRIORITY 3 -> invest in resilience
PRIORITY 4 -> preparedness response and Build Back Better
RULE -> fields of action, not the seven global targets

ASCII MASTER FLOW - PANEL 9/12: Seven-target logic

TARGETS A TO D -> substantially reduce specified losses
MORTALITY AND AFFECTED PEOPLE -> human impact directions
ECONOMIC AND CRITICAL-SERVICE LOSS -> system impact directions
TARGETS E TO G -> substantially increase strategies cooperation warnings
RULE -> indicator values require official monitoring metadata

ASCII MASTER FLOW - PANEL 10/12: Global-domestic bridge

SENDAI -> global principles priorities and targets
INDIAN LAW -> authorities powers duties and funds
NATIONAL AND STATE PLANS -> domestic implementation choices
SECTOR AGENCIES -> warnings infrastructure health and relief
ALIGNMENT -> policy bridge, not legal identity
CLOSE DISTINCTION: Hazard is not disaster, resilience is not mere recovery, NDMA is not...
CLOSE DISTINCTION: Hazard is not disaster, resilience is not mere recovery, NDMA is not...

ASCII MASTER FLOW - PANEL 11/12: Evidence status screen

LOSS FIGURE -> event date definition and reporting source
TARGET PROGRESS -> indicator baseline and period
EARLY WARNING -> hazard geography reach and access
INSTITUTIONAL POWER -> current statutory provision
PLAN STATUS -> dated official edition or amendment
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Preserve dedicated Disaster Management...
EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Preserve dedicated Disaster Management...

ASCII MASTER FLOW - PANEL 12/12: Disaster answer spine

DEFINE -> hazard exposure vulnerability capacity risk and disaster
SEQUENCE -> prevention mitigation preparedness response recovery
MAP -> NDMA NEC NIDM NDRF SDMA and DDMA by owned role
SEPARATE -> four Sendai priorities from seven global targets
VERIFY -> domestic law plan loss indicator and warning evidence